



CHEM 1008 TA Tutorial

Chapter 7: Electrons in Atoms and the Periodic Table

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2024.12.06
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Fireworks and flame colours



東風夜放花千樹，更吹落、星如雨。-- 辛棄疾《青玉案·元夕》

The east wind scatters a thousand blossoms in the night,
And blows down stars, falling like rain.



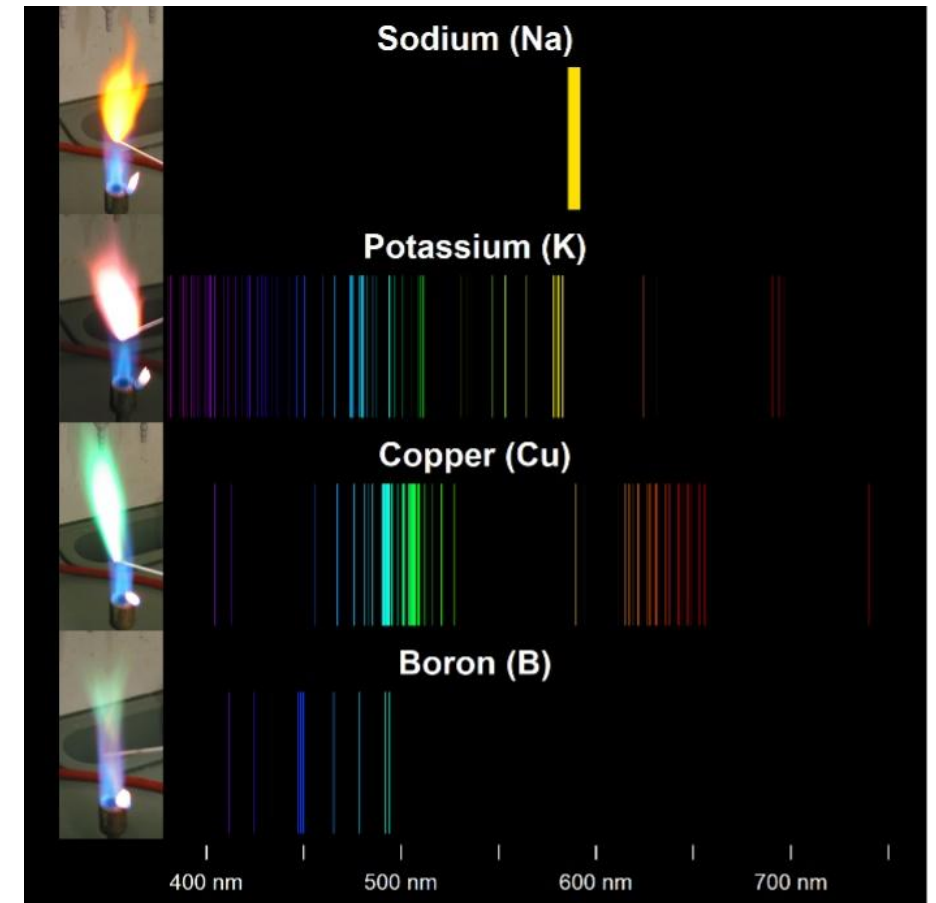
李旼(LI Tian, 621~?): The first person
who made firework

Element Spectrum: A way to find ores



In the 19th century, with the development of the Industrial Revolution, the demand for metallurgy and mineral screening increased sharply. The emission spectra of elements became an important method for identifying minerals.

Attain more knowledge about the spectrum of the element means “money”!



Element Spectrum: A way to find ores

The found metal elements from the spectrum:

铯 (Cs, Caesium): **Blue line**, name from *Caesius* (Latin), blue sky, 1860

铷 (Rb, Rubidium): **Red line**, name from *Rubidus* (Latin), deep red, 1861

铊 (Tl, Thallium): **Green line**, name from *Thallos* (Greece), green branches, 1861

铟 (In, Indium): **Blue line**, name from *Indigo*, 1863

镓 (Ga, Gallium): **Purple line**, 1875

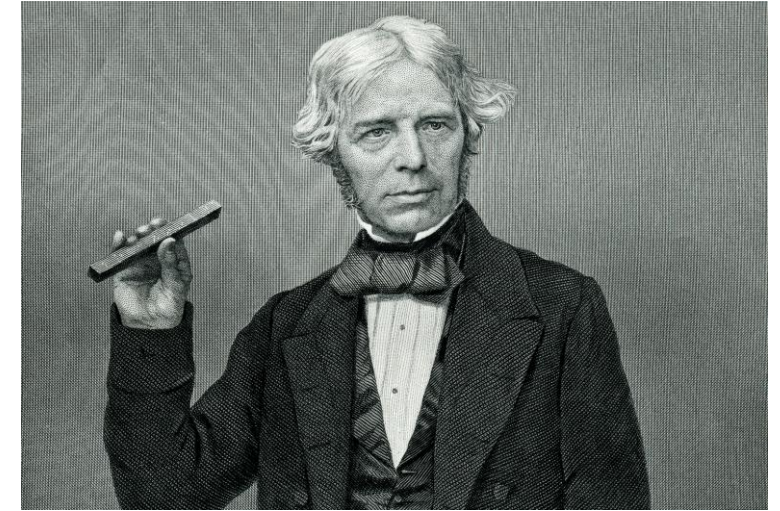
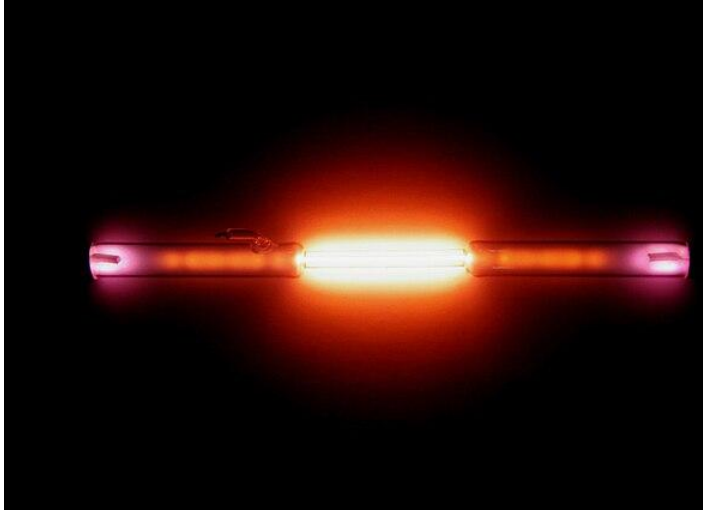
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Why do atoms produce the light?

How to explore the mechanism by suitable research model?

Development of Gas-discharge lamps



Gas-discharge lamps

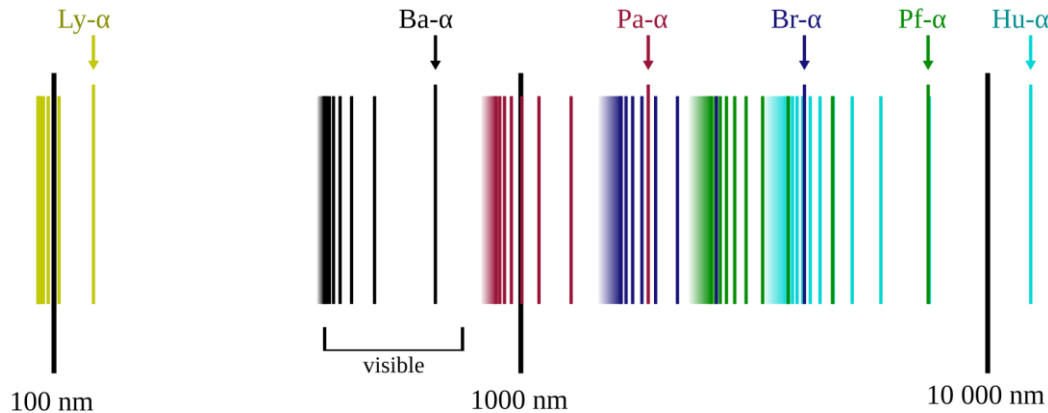
Michael Faraday

In the early 18th century, the **first gas-discharge tube** was invented. In the early 19th century, **Faraday** studied the **relationship between electricity and gases**. After a century of development, the spectrum of gases glowing under electrical excitation was thoroughly studied.

Gas-discharge tube offer the opportunity.

Gas light spectrum is relatively simple.

The hydrogen light spectrum



Hydrogen light spectrum



Johann Jakob Balmer

Balmer build an empirical formula for the hydrogen atom spectrum. The formula is named by “*Balmer formula*” (1884)

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (n_2 > n_1)$$

Hydrogen light spectrum has 6 different series:

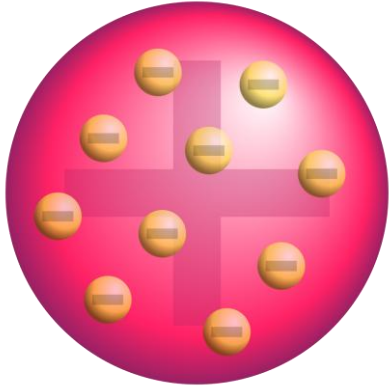
Lyman series, *Balmer* series (visible spectrum), *Paschen* series, *Brackett* series, *Pfund* series, *Humphreys* series

Rydberg formula (1888), uniform formula

$$\frac{1}{\lambda} = Z^2 R_{\infty} \left(\frac{1}{n'^2} - \frac{1}{n^2} \right)$$

Problem: Why do they not emit a continuous spectrum?

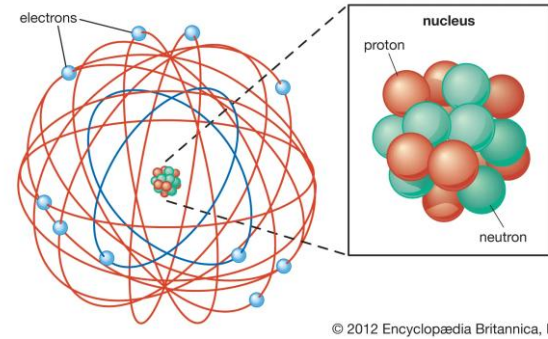
Failure Atom Model



Plum pudding model (1904)

Assume: The electron is embedded into the Proton. The electron is static.

- How does atom produce light?



Planet Model (1911)

Assume: The electrons are rotating around the nucleus. Electrons' motion produces light

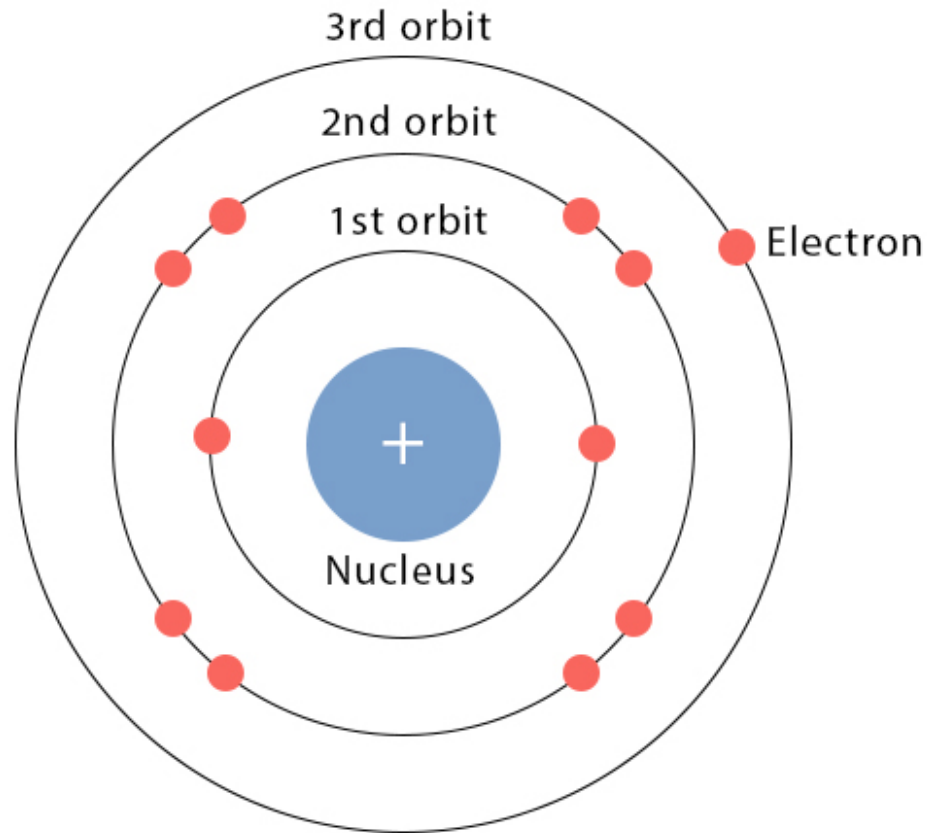
- Why do they not emit a continuous spectrum?

Failure of classical physics:

If the electron is free move at any place in atom, it cannot explain the line spectrum

Bohr's Model

Bohr Model (1913)



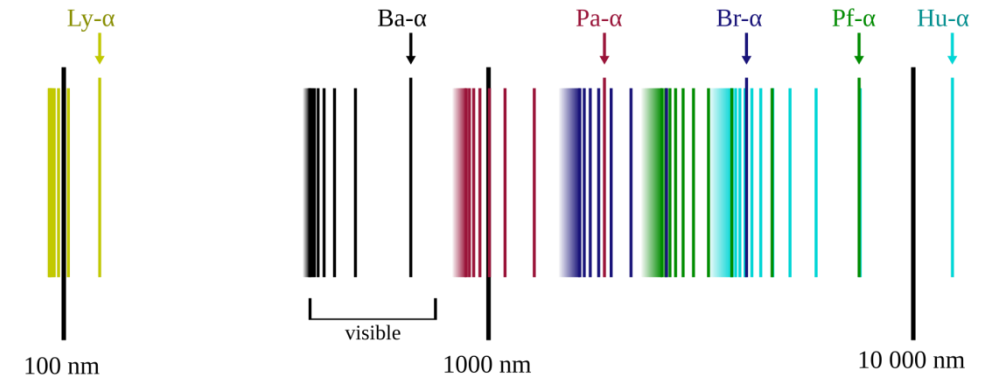
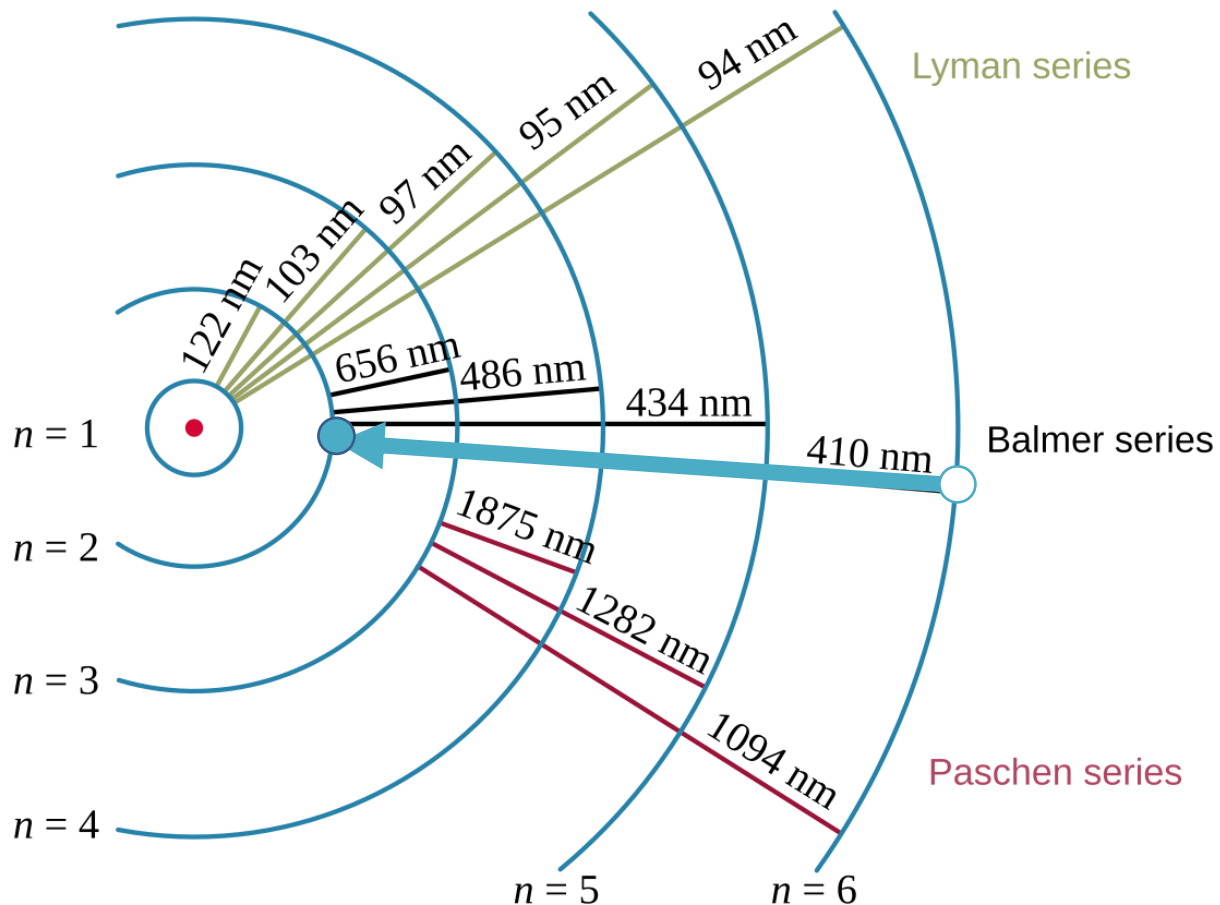
Neil Bohr

Electrons only move in a fixed orbital. Between the different orbitals, the electron only can “Jump”. “Jump” produces light.

The general rule is for the n^{th} shell, it can hold up to $2n^2$ electrons.

$n = 1$	->	2
$n = 2$	->	8
$n = 3$	->	18
$n = 4$	->	32
$n = 5$	->	50
$n = 6$	->	72
$n = 7$	->	98

Bohr's Model



The different hydrogen light spectrum series can be considered the “jump” (transition) among different orbitals. For example, the Balmer series is from the transition among high energy orbits and $n = 2$ orbit

From the two formula,

$$E_n = \frac{-13.6 \text{ eV}}{n^2} \quad \Delta E = h\nu = \frac{hc}{\lambda}$$

We can easily calculate the wavelength of emission light

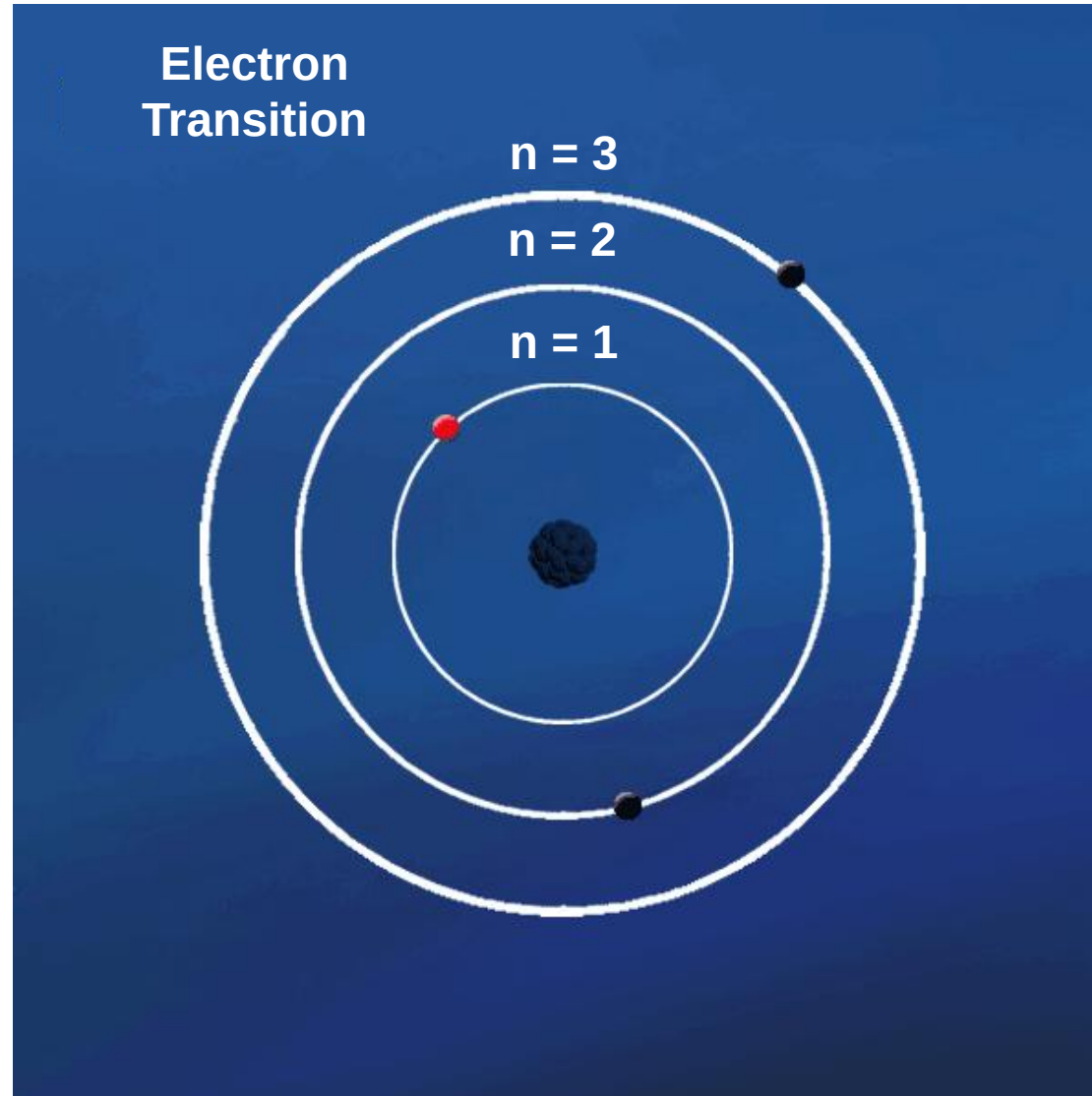
The Electron Transition in Bohr's Model

Electronic Excitation

From low energy level to high energy level

Absorb energy

Electron Transition



Electronic Relaxation

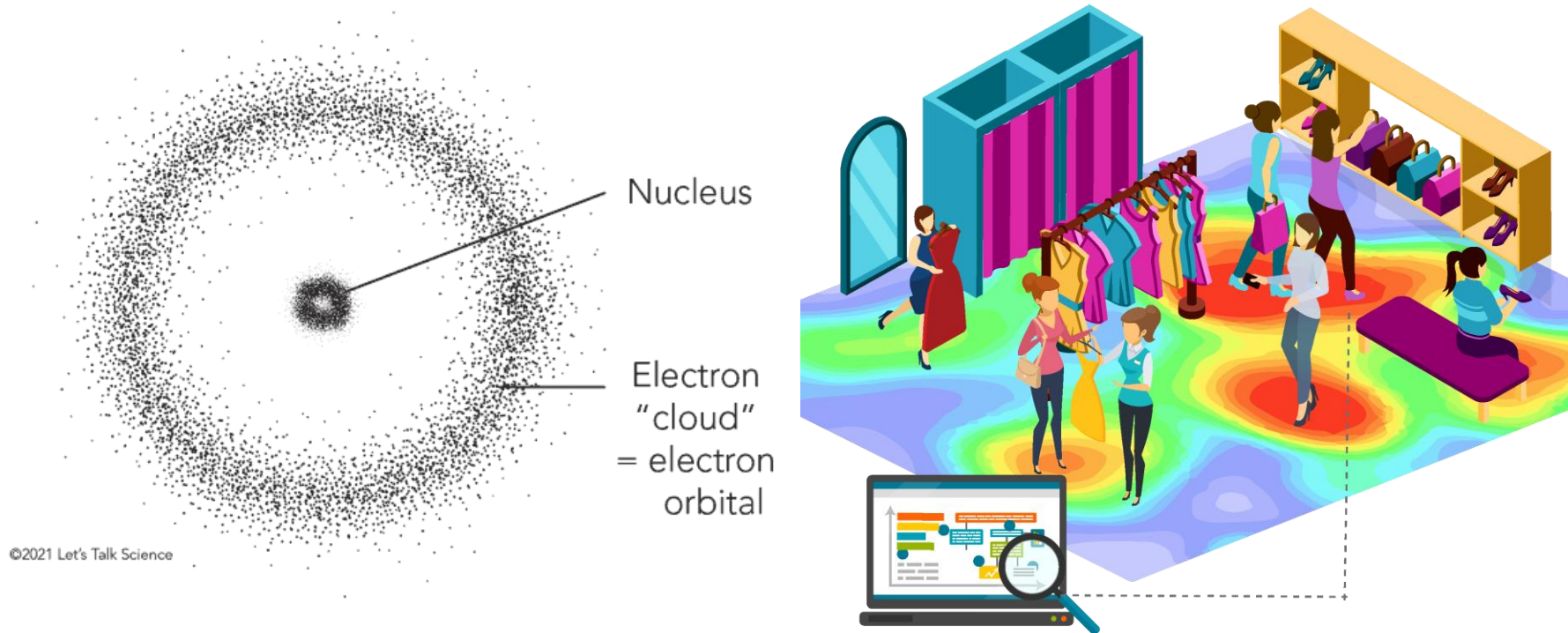
From high energy level to low energy level

Emit light

Electron Cloud Model

In Bohr's model, electron motion is in fixed orbitals.

However, the electron randomly moves. But most likely electrons move in the fixed area.



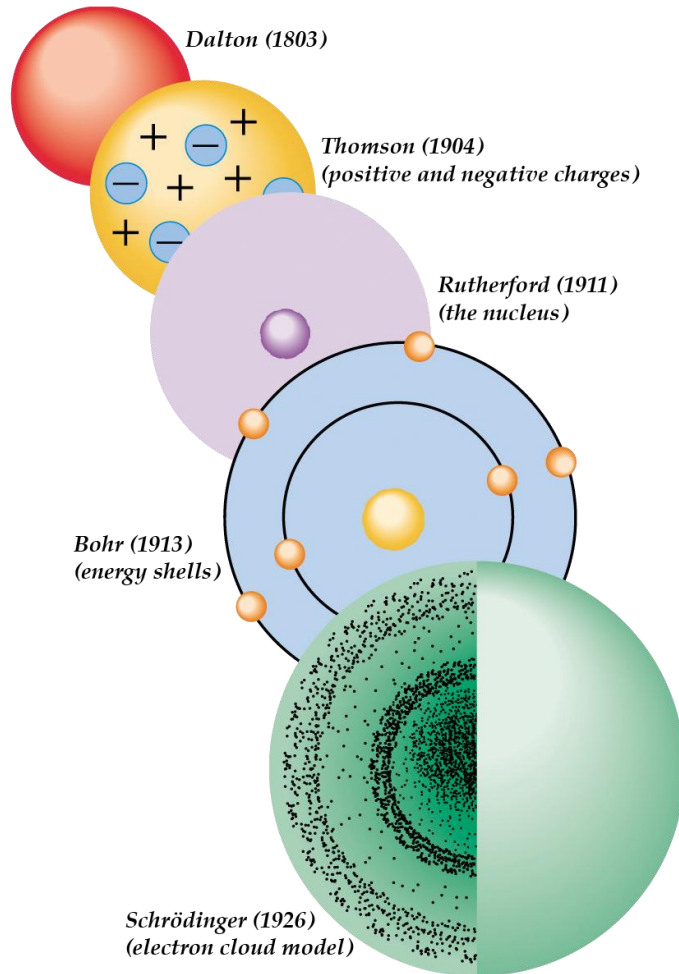
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Electronic motion is similar to the people in the shop.

Although people (electrons) randomly move, people (electrons) usually move to some specific area.

This area is similar to a “cloud shell”. The shell is equal to the “orbital” in “Bohr's model”

The Development of Atom Model



- 1803: **Dalton** thought the matter is build on atom. Atom is the least unit of matter;
- 1904: After the found of electron, **Thomson** assume a atom model. It's similar with “**Plum pudding**”, Electron is embedded into the proton;
- 1911: The **Rutherford** built the atom model based on the “**Planet**”;
- 1913: The **Bohr** assume the electron orbits is fixed value;
- 1926: The **Schrodinger** assume the electron motion is a “**Could**”;

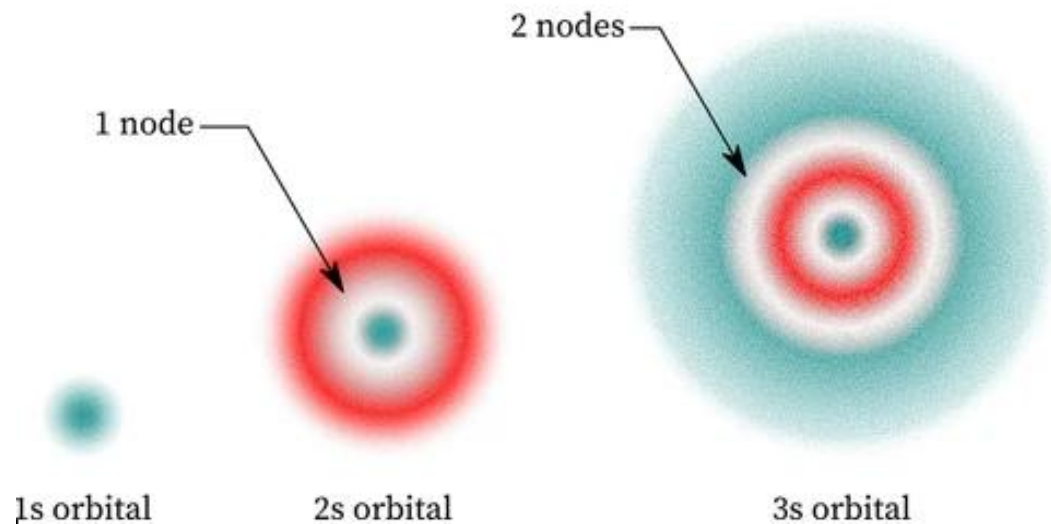
What is pushing the development of the atom model?

The needs of industry and capital

Electron Cloud Model: How to Describe Electron

Principal Quantum Number (n): Shell number Bohr's Model (1913)

Angular Momentum Quantum Number (l): Cloud shape Sommerfeld Model (1916)

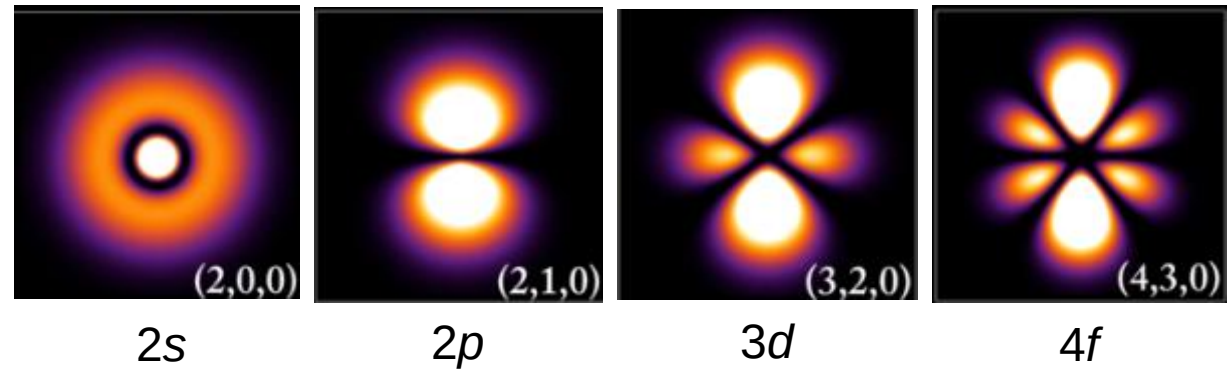


$l = 0$, s, 1 subshell, sphere, 2 electron

$l = 1$, p, 3 subshells, dumbbell, 6 electron

$l = 2$, d, 5 subshells, four-leaf clover or circular, 10 electron

$l = 3$, f, 7 subshells, more complex, 14 electron

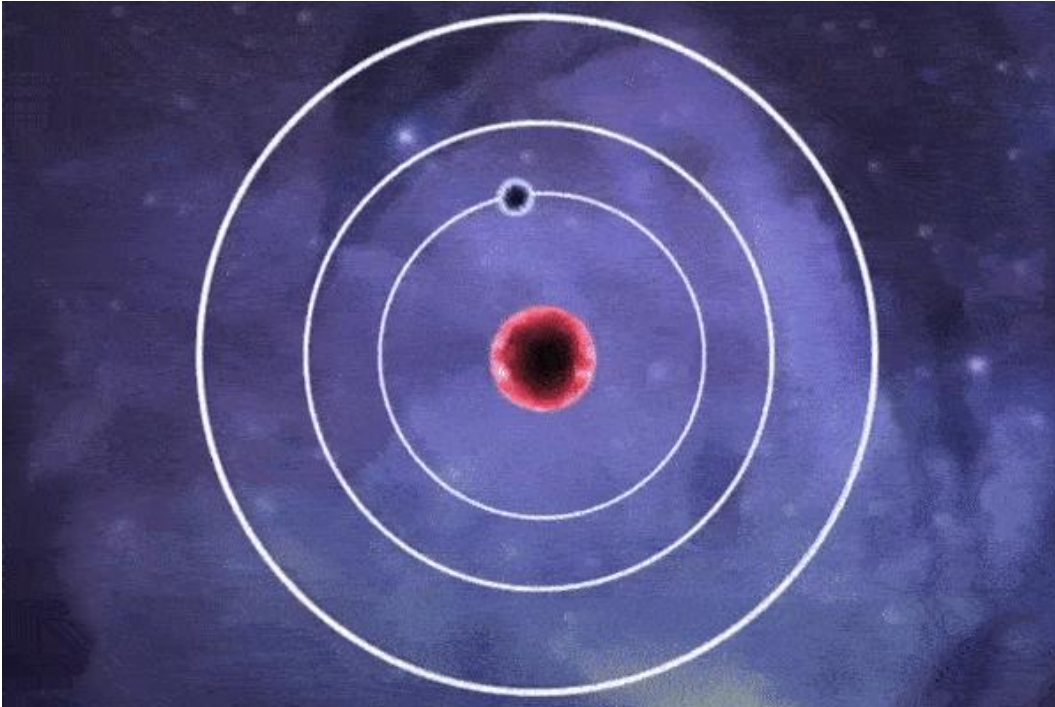


Electron Cloud Model: How to Describe Electron

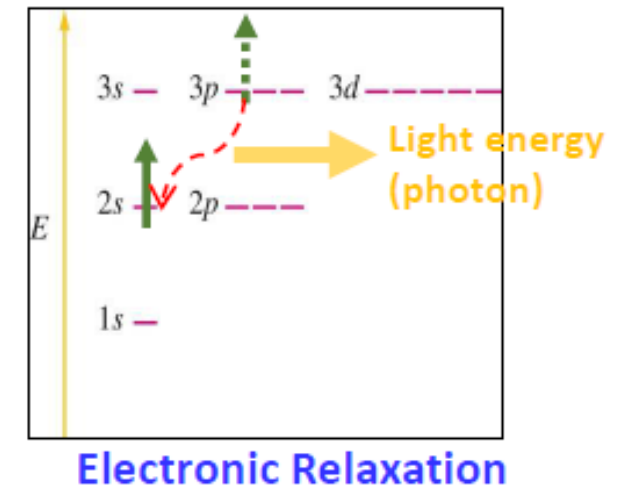
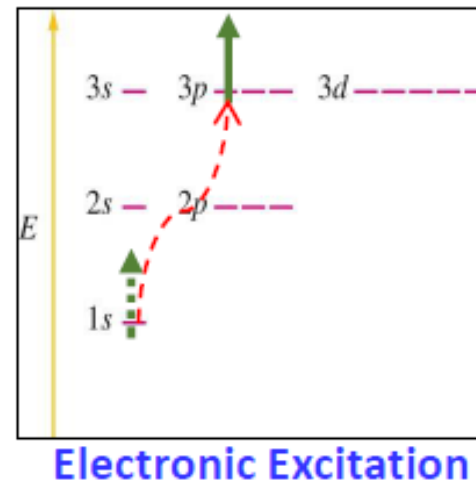
	s ($\ell = 0$)	p ($\ell = 1$)			d ($\ell = 2$)					f ($\ell = 3$)							
	$m = 0$	$m = 0$	$m = \pm 1$		$m = 0$	$m = \pm 1$		$m = \pm 2$		$m = 0$	$m = \pm 1$		$m = \pm 2$		$m = \pm 3$		
	s	p_z	p_x	p_y	d_{z²}	d_{xz}	d_{yz}	d_{xy}	d_{x²-y²}	f_{z³}	f_{xz²}	f_{yz²}	f_{xyz}	f_{z(x²-y²)}	f_{x(x²-3y²)}	f_{y(3x²-y²)}	
$n = 1$		1s ²															2
$n = 2$					2s ² 2p ⁶												8
$n = 3$										3s ² 3p ⁶ 3d ¹⁰							18
$n = 4$																	32
$n = 5$										...	...	...	...	...	...	...	...
$n = 6$					... †	... †	... †	... †	... †	... *	... *	... *	... *	... *	... *	... *	...
$n = 7$		... †	... †	... †	... *	... *	... *	... *	... *	... *	... *	... *	... *	... *	... *	... *	...

The Electron Transition in “Electron Cloud”

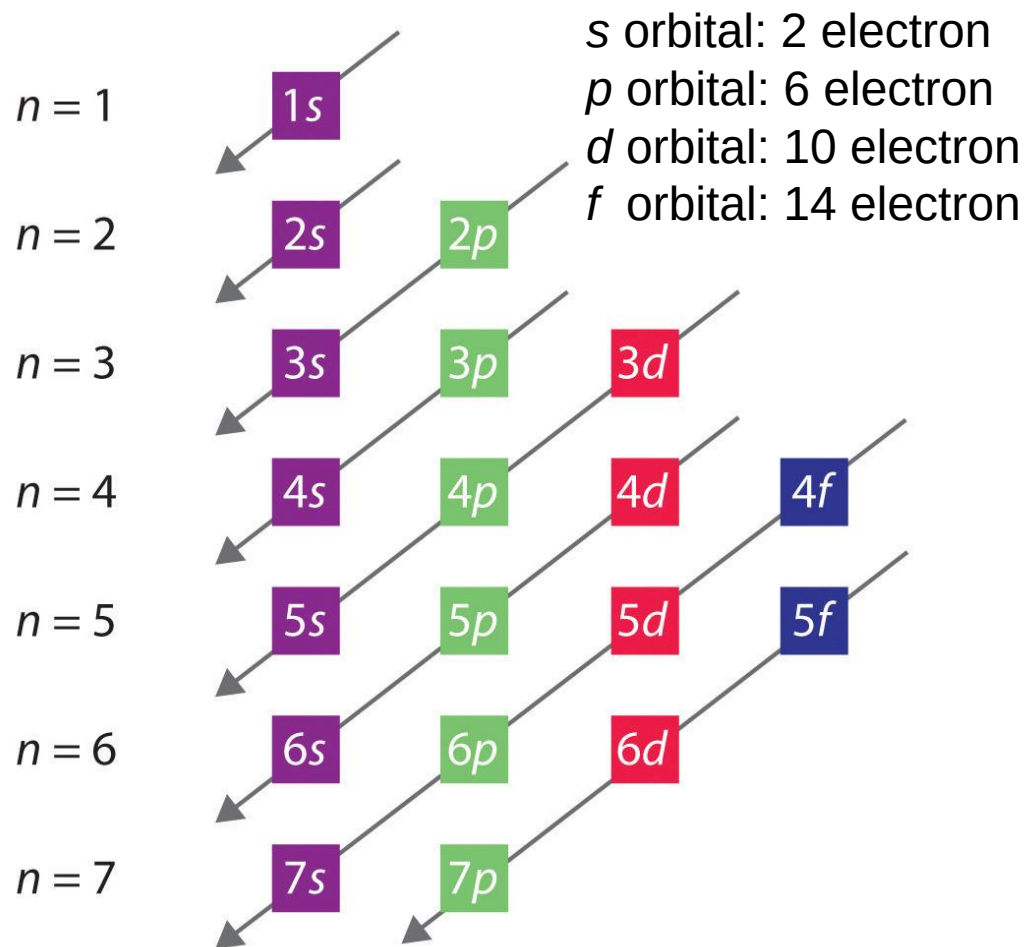
The electron transition in “Bohr’s Model”



The electron transition in “Electron Cloud”



Electronic Configuration



Example:

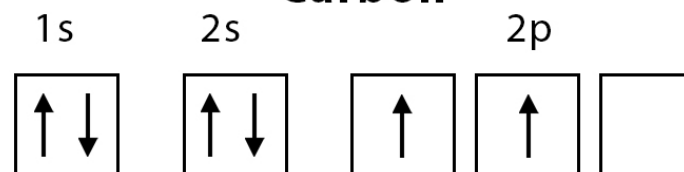
	<u>Z (# e-)</u>	<u>Electronic configurations</u>	<u>Orbital Diagram</u>
H	1	$1s^1$	1s ↑ 2s 2p
He	2	$1s^2$	1s ↑↓ 2s 2p

Pauli Exclusion Principle: states that orbitals can hold no more than two electrons (with opposite spins).

Hund's Rule: when filling orbitals of equal energy, electrons fill them singly first, with parallel spins.

Hund's Rule Examples

Carbon

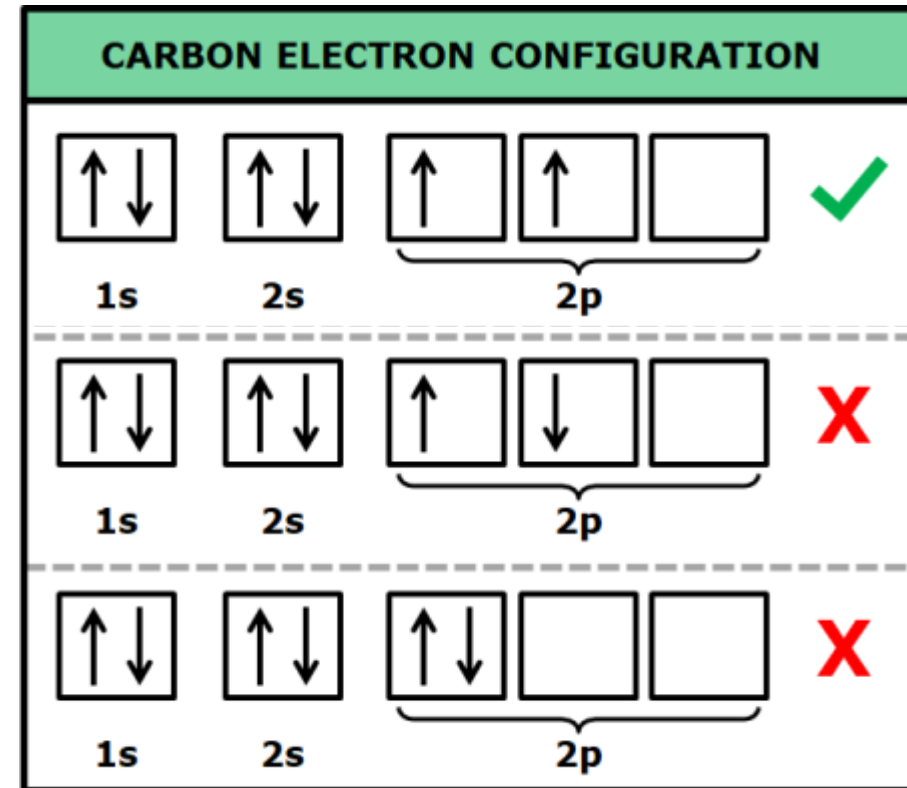
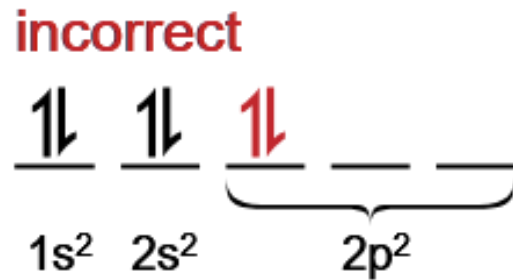
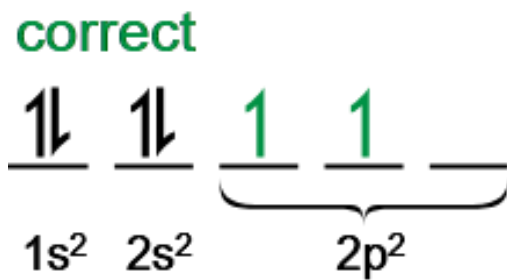
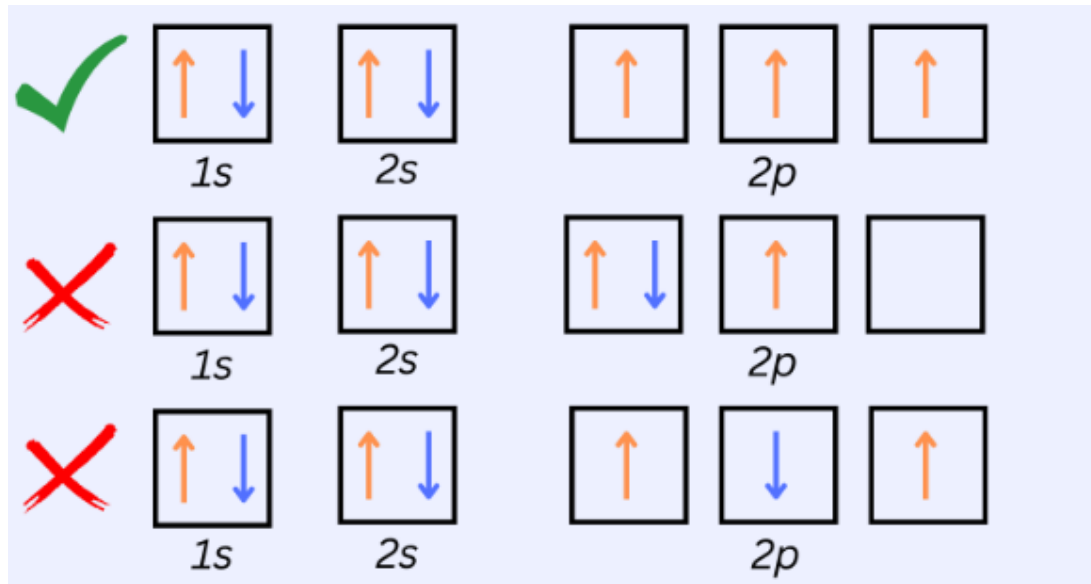


Nitrogen



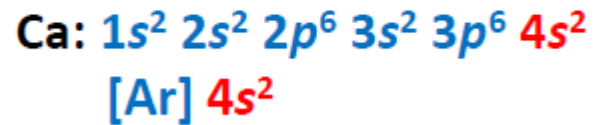
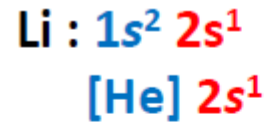
ChemistryLearner.com

More examples

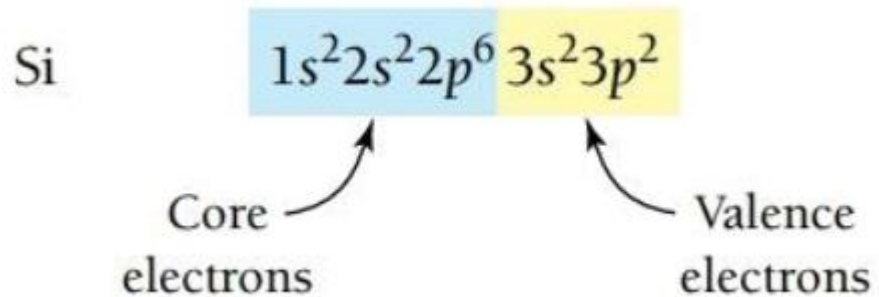


Abbreviated Form and Valence Electrons

The core electrons are represented using the symbol of the preceding noble gas.



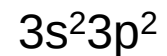
Valence electrons: electrons in the outermost principal shell (the principal shell with the highest n value).



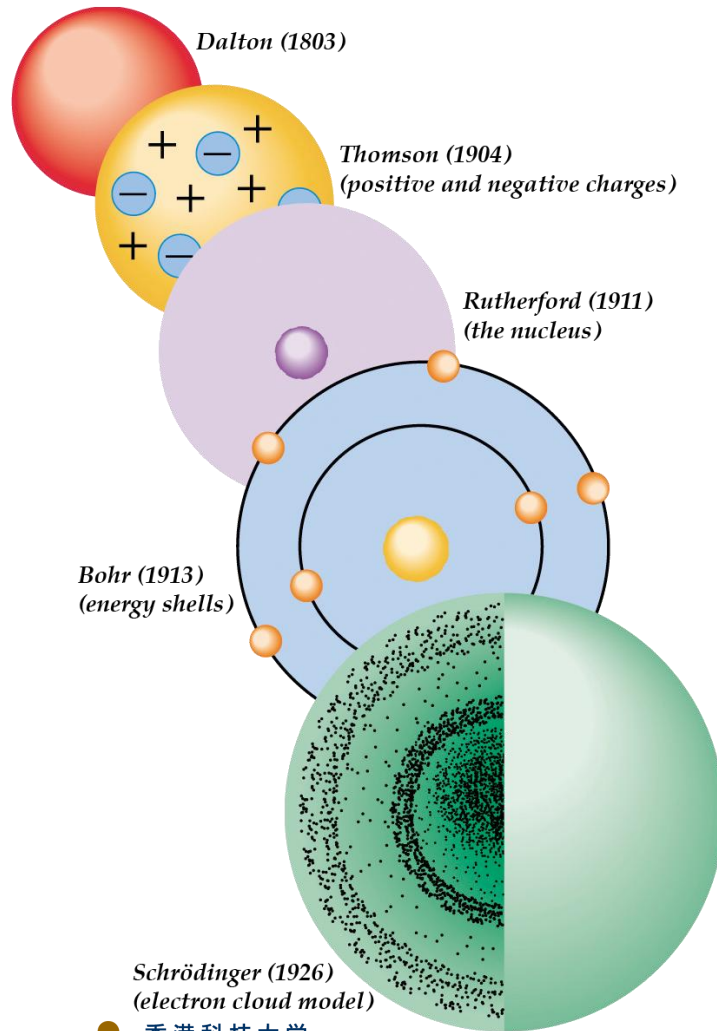
Abbreviated Form:



Valence Electrons:



Conclusion



SOLVAY CONFERENCE 1927

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	A. PICARD	E. HENRIOT	P. EHRENFEST	Ed. HERSEN	Th. DE DONDER	E. SCHRÖDINGER	E. VERSCHAFFELT	W. PAULI	W. HEISENBERG	R.H FOWLER	L. BRILLOUIN
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Absents : Sir W.H. BRAGG, H. DESLANDRES et E. VAN AUBEL

1927 5th Solvay Conferences “*Electrons and Photons*”